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Parenting Practices in Families of Children With Comorbid ADHD and Anxiety Disorders: A Systematic Review and Meta-Analysis

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ABSTRACT

Background: Comorbid anxiety disorders affect up to 28% of children with attention-deficit/hyperactivity disorder (ADHD). While coercive parenting is often linked to ADHD and overprotectiveness to anxiety disorders (ADs), little is known about parenting practices in children with both disorders (ADHD+ADs). This systematic review and meta-analysis examined parenting practices in children aged 5–12 years with ADHD+ADs.

Methods: Studies were identified through PsycINFO (Ovid), Medline (Ovid), EMBASE, and Web of Science from inception date to January 23, 2025. Of the 1838 records screened, four studies were included in the review, and three were included in the meta-analysis.

Results: Compared to parents of children with ADHD alone, those of children with ADHD+ADs reported greater use of anxiety-related parenting practices ($k = 3$, $g = 0.32$, 95% CI = 0.11, 0.53, $p = 0.002$) and fewer positive parenting practices ($k = 3$, $g = -0.35$, 95% CI = −0.55, −0.14, $p = 0.001$). No significant differences were found in ADHD-related parenting practices ($k = 2$, $g = 0.27$, 95% CI = −0.16, 0.70, $p = 0.26$).

Conclusion: These findings suggest that families of children with ADHD+ADs may be characterized by increased anxiety-related parenting practices and reduced positive parenting compared to families of children with ADHD alone. Further research is needed to inform the development of targeted interventions.

1 | Introduction

Approximately 28% of children diagnosed with attention-deficit/hyperactivity disorder (ADHD) also meet criteria for a comorbid anxiety disorder (Shroff and Sanchez-Lacay 2018). This co-occurrence is associated with more severe consequences than either disorder alone (Bowen et al. 2008; Mikami et al. 2011; Shroff and Sanchez-Lacay 2018; Souza et al. 2005). For instance, children with comorbid ADHD and anxiety disorders (ADHD+ADs) exhibit greater school-related worries, higher rates of absenteeism (Sciberras et al. 2014) and an elevated risk of academic failure (Jensen et al. 2001) compared to those with only one of these disorders. They also present more significant deficits in social skills and are more vulnerable to peer rejection and isolation (Mikami et al. 2011; Nadeau et al. 2018). Furthermore, children with ADHD+AD have lower self-esteem, reduced quality of life and a greater risk of suicidal ideation than children with either ADHD or anxiety disorders (Choi et al. 2017; Sciberras et al. 2014; Tannock 2000; Tannock 2009). At the family level, this comorbidity is linked

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Summary

- This systematic review identified only four studies examining parenting practices in children with ADHD+ADs.
- Families of children with ADHD+ADs exhibit more anxiety-related parenting practices and lower levels of positive parenting practices compared to families of children with ADHD alone.
- The knowledge gap regarding parenting practices in these families is concerning given their importance in evidence-based interventions for both ADHD and anxiety disorders.
- These findings highlight the potential clinical value of targeting disorder-specific parenting behaviours while enhancing positive parenting practices to improve outcomes for children with ADHD+ADs and their families.

to increased relational strain (Chan et al. 2022; Tannock et al. 1995) and poorer parental mental health (Eun et al. 2018; Mulraney et al. 2018; Pfiffner and McBurnett 2006; Sciberras et al. 2014; Shroff and Sanchez-Lacay 2018).

Parenting practices play a central role in the maintenance and exacerbation of both ADHD and anxiety disorders in children (Chronis-Tuscano et al. 2008; Claussen et al. 2024; Vasey et al. 2014). For instance, coercive and aggressive parenting is more commonly associated with ADHD (Chronis et al. 2004; Chronis-Tuscano et al. 2008; Cussen et al. 2012; Gair et al. 2021; Keown and Woodward 2002; Pfiffner et al. 2005; Woodward et al. 1998), whereas overprotectiveness, excessive control and reinforcement of avoidance behaviours are more specifically linked to anxiety disorders (Drake and Ginsburg 2012; Laurin et al. 2015; McLeod et al. 2007; Murray et al. 2009; Negreiros and Miller 2014; van der Bruggen et al. 2008; Vasey et al. 2014). The literature suggests parenting practices typically associated with each disorder are overrepresented in families of children with ADHD+ADs (Kepley and Ostrander 2007; Pfiffner and McBurnett 2006; Yap and Jorm 2015). Furthermore, this comorbidity has also been linked to a reduction in positive parenting practices (e.g., warmth, communication, involvement, positive discipline; Gair et al. 2021; Kepley and Ostrander 2007; Pfiffner and McBurnett 2006, Yap and Jorm 2015), which otherwise serve as protective factors (Clayborne et al. 2021; Haack et al. 2016; McRae et al. 2020).

Despite the well-established clinical and theoretical relevance of parenting practices in the course of ADHD and anxiety disorders (Claussen et al. 2024; Coates et al. 2015; Turgeon et al. 2023; Zwi et al. 2011), few studies have specifically examined their association with ADHD+ADs. To our knowledge, no systematic review or meta-analysis has been conducted on this topic. Advancing our understanding of parenting practices associated with ADHD+ADs is essential to inform the development of more effective psychosocial interventions. Indeed, parent training is a core component of ADHD management (Wolraich et al. 2019), and family-based interventions are increasingly recognized as effective in treating childhood anxiety disorders (Byrne et al. 2023; Schwartz et al. 2019), especially in the presence of comorbidity such as ADHD (van Steensel et al. 2024).

2 | Objectives

This systematic review and meta-analysis aimed to identify parenting practices associated with ADHD+ADs in children aged 5–12 years, and, when possible, to compare them with those observed in children with either condition alone.

3 | Method

This systematic review and meta-analysis were conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al. 2021). The protocol was registered in PROSPERO (CRD42021219479).

3.1 | Search Strategy

Search strategies were developed in collaboration with an information specialist and adapted for each database (see Supplementary Table 1). Studies were identified through PsycINFO (Ovid), Medline (Ovid), EMBASE and Web of Science from inception date to January 23, 2025. The ProQuest Dissertations & Theses Global database was also consulted to identify unpublished literature. Additionally, reference lists of included studies were manually screened.

3.2 | Eligibility Criteria

Studies were eligible if they (1) included children aged 5–12 years diagnosed with ADHD+ADs, (2) assessed parenting practices and (3) were published in English or French. Studies were excluded if the sample comprised children with autism spectrum disorder or psychotic disorders.

3.3 | Study Selection

Two independent reviewers screened studies using the Covidence Systematic Review Software (Veritas Health Innovation 2020). Discrepancies were resolved through discussion, and when consensus could not be reached, a third reviewer (ID) was consulted. A total of 2656 studies were identified, and after removing the 371 duplicates, 26 full-text articles were assessed. Four studies met the inclusion criteria, including one identified through reference lists. See the PRISMA flow diagram in Figure 1 for details.

3.4 | Data Extraction and Risk of Bias Assessment

Data extraction and risk of bias assessment were independently conducted by two reviewers. Any discrepancies

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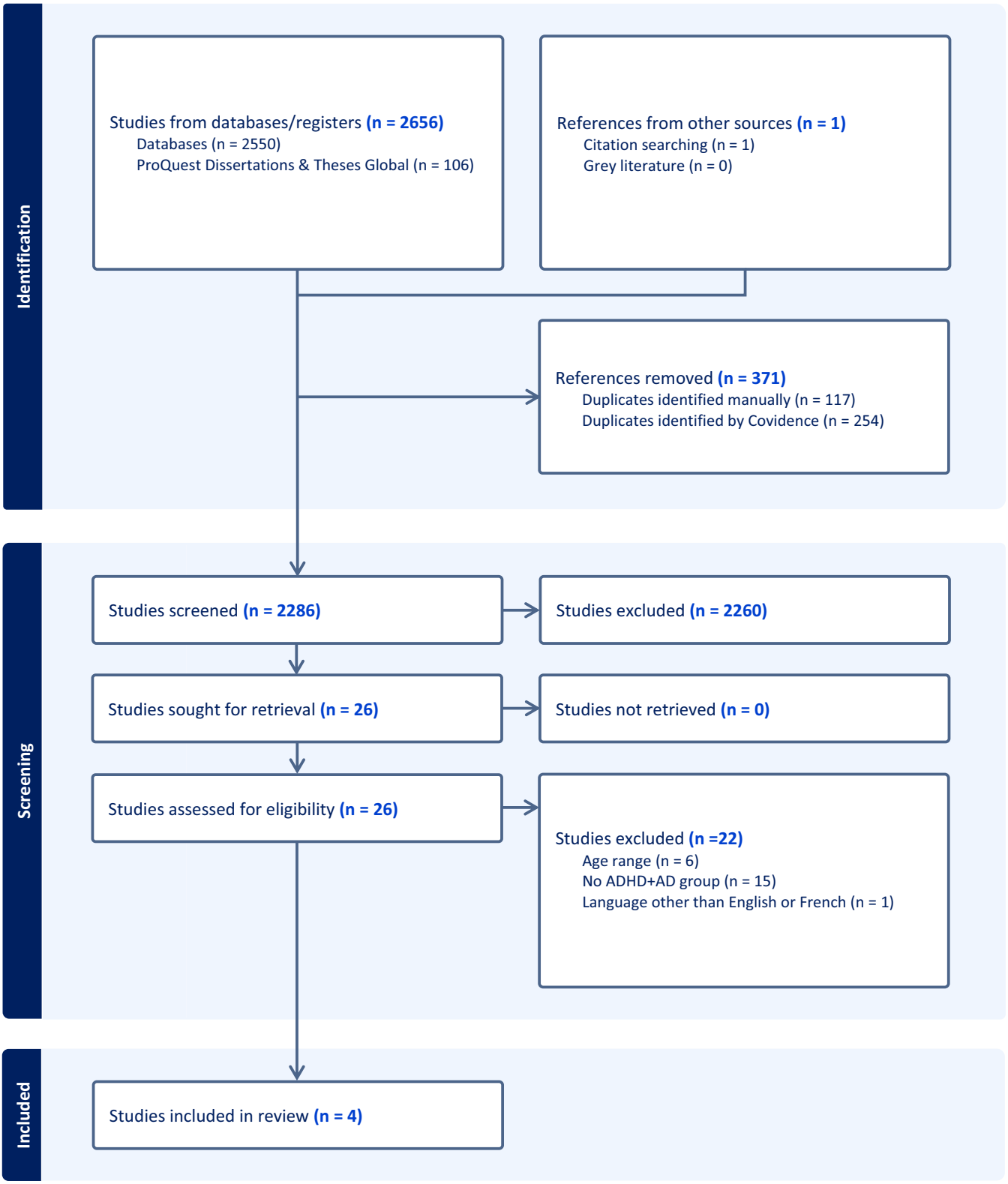


FIGURE 1 | PRISMA study selection flowchart.

were resolved using the same procedure as for study selection. Extracted data included participant characteristics, study design, diagnostic procedures, parenting measures and results (including effect sizes when available). Authors of three studies were contacted to obtain missing data; one study was excluded from the meta-analysis due to unavailable effect size data (McGee 2007). Risk of bias was assessed using the Quality Appraisal Checklist—Quantitative Intervention Studies (National Institute for Health and Care Excellence [NICE], 2012).

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3.5 | Data Analysis

Parenting practices were categorized as follows, based on the literature: (1) parenting practices typically associated with ADHD (e.g., coercive and aggressive discipline; Chronis et al. 2004; Chronis-Tuscano et al. 2008; Cussen et al. 2012; Keown and Woodward 2002; Pfiffner et al. 2005; Woodward et al. 1998), (2) those linked with anxiety disorders (Drake and Ginsburg 2012; Laurin et al. 2015; McLeod et al. 2007; Murray et al. 2009; Negreiros and Miller 2014; van der Bruggen et al. 2008; Vasey et al. 2014) and (3) positive parenting practices (see Table 1 for details).

Effect sizes were estimated using Hedge's *g*, as it provides greater accuracy than Cohen's *d* in small samples (Borenstein et al. 2009). A random-effects model was applied to account for methodological heterogeneity across studies (Borenstein et al. 2009). When multiple parenting practices within a category were assessed in a single study, their mean effect size was computed to represent that category. Heterogeneity was examined using the *Q* statistic; if statistically significant ($p < 0.05$), the proportion of variance attributable to heterogeneity was calculated using the I^2 statistic (Higgins and Thompson 2002) and interpreted as follows: 25% = low, 50% = moderate and 75% = high (Borenstein et al. 2009; Card 2015). Subgroup analyses were planned to explore heterogeneity when I^2 reached 40% or more (Deeks et al. 2024) and to examine the influence of study characteristics on findings. However, these analyses could not be conducted due to the small number of eligible studies. All analyses were performed using Comprehensive Meta-Analysis Software (Biostat Inc., Version 3). Quantitative analyses were complemented by a descriptive synthesis of individual study results.

4 | Results

4.1 | Characteristics of the Studies

The four studies included in this review were published between 2006 and 2013. Two were peer-reviewed journal articles and two were doctoral dissertations. Three studies used cross-sectional designs (Kepley and Ostrander 2007; McGee 2007; Pfiffner and McBurnett 2006) and one used a longitudinal design (Curchack-Lichtin 2013). In the latter study, only the assessment conducted at age 6 provided data on ADHD, anxiety disorders and parenting practices; this time point was therefore included in the meta-analysis. One study provided effect size data exclusively for statistically significant results (Pfiffner and McBurnett 2006), and another did not provide sufficient data for effect size calculation (McGee 2007).

A total of 122 children with ADHD+ADs and 278 with ADHD only were included in the four selected studies. None of the studies included a comparison group of children with anxiety disorders only. All participants were between 5 and 12 years of age and resided in the United States. Demographic data indicated that the samples were predominantly composed of Caucasian children (73.18%, $n = 221/302$, 3 studies) and boys (75.17%, $n = 227/302$, 3 studies). The mean age, reported in two studies, was 8.36 years (SD = 1.75). One study described the children as being school age (Kepley and Ostrander 2007). Parenting practices were assessed with one or more of the following instruments: Alabama Parenting Questionnaire (APQ; Shelton et al. 1996), Alabama Parenting Questionnaire—Preschool Revision (APQ-PR; Clerkin et al. 2007), Behavioural Assessment System for Children—Parent Personality Profile (BASC-PPP; Reynolds and Kamphaus 1992), Family Environment Scale (FES; Moos and Moos 1981), Parent–Child Relationship Scale—Brief Version (PCRS; Furman and Giberson 1995) or an observational measure of parent–child interactions (Curchack-Lichtin 2013). Study characteristics are presented in Table 2.

4.2 | Risk of Bias

Due to the limited number of studies included in this review, a funnel plot could not be generated to visually assess the potential for publication bias. As summarized in Table 3, the quality appraisal indicated that most studies adequately addressed or controlled for key sources of bias. However, reporting on eligibility

TABLE 1 | Categories of parental practices.

Parenting practices typically associated with ADHD	Parenting practices typically associated with anxiety disorders	Positive parenting practices
APQ: • Inconsistent discipline • Poor monitoring/supervision • Corporal punishment APQ-PR: • Inconsistent discipline • Punitive parenting FES: Conflict PCRS: Power assertion PCI: Parent negative affect (anger or rejection of the child)	BASC-PPP: Parental control FES: • Achievement orientation • Control • Independence • Organisation PCI: • Quality of assistance • Respect for autonomy PCRS: Possessiveness (overprotection)	APQ: • Positive parenting • Involvement APQ-PR: Positive parenting BASC-PPP: Communication/ Involvement FES: • Active-recreational orientation • Cohesion • Expressiveness PCRS: Warmth PCI: Emotionally supportive presence

Abbreviations: APQ, Alabama Parenting Questionnaire; APQ-PR, Alabama Parenting Questionnaire—Preschool Revision; BASC-PPP, Behavioural Assessment System for Children—Parent Personality Profile; FES, Family Environment Scale; PCI, observer recording of parent–child interactions; PCRS, Parent–Child Relationship Scale—Brief Version.

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TABLE 2 | Study characteristics.

Studies	Groups	Sex (%)	Age (mean, SD)	Race (%)	Measures
Curchack-Lichtin (2013)	ADHD+AD: $n=8$ ADHD: $n=88$	Girls: $n=25$ (26) Boys: $n=71$ (74)	6[a]	Caucasians (60) African American (12) Asian descent (11) Mixed racial background (19)	• Alabama Parenting Questionnaire, Preschool Revision (APQ-PR; Clerkin et al. 2007) • Observer recording of parent–child interactions (Curchack-Lichtin 2013)
Kepley and Ostrander (2007)	ADHD+AD: $n=18$ ADHD: $n=80$	NA	School age (Grades 1–4)	NA	• Behavioural Assessment System for Children—Parent Personality Profile (BASC-PPP; Reynolds and Kamphaus 1992) • Family Environment Scale (FES; Moos and Moos 1981)
McGee (2007)	ADHD+AD: $n=31$ ADHD: $n=32$	Girls: $n=25$ (40) Boys: $n=38$ (60)	9.43 (1.41)	Caucasians (76.2) African Americans (20.6) Biracial (3.2)	• Alabama Parenting Questionnaire (APQ; Shelton et al. 1996)
Pfiffner and McBurnett (2006)	ADHD+AD: $n=64$ ADHD: $n=79$	ADHD+AD: Girls: $n=12$ (19) Boys: $n=52$ (81) ADHD: Girls: $n=13$ (16) Boys: $n=66$ (84)	ADHD+AD: 7.8 (1.7) ADHD: 8.1 (1.7)	Caucasians (79) Mixed origin (11) Other (< 5)	• Alabama Parenting Questionnaire (APQ; Shelton et al. 1996) • Parent–Child Relationship Scale—Brief Version (PCRS; Furman and Giberson 1995)

Abbreviation: NA, not available.
[a]The children were 6 years old at the selected time point from this longitudinal study.

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TABLE 3 | Risk of bias assessed with the Quality Appraisal Checklist for Quantitative Intervention Studies (National Institute for Health and Care Excellence [NICE], 2012).

Studies	Criteria								
	Population	Sample	Eligibility	Measures	Results	Effect size	Statistical analysis	Internal validity	External validity
Curchack-Lichtin (2013)	++	+	+	++	++	++	–	+	+
Kepley and Ostrander (2007)	–	+	+	++	+	++	++	–	+
McGee (2007)	++	+	–	++	++	+	++	++	+
Pfiffner and McBurnett (2006)	++	+	+	++	++	++	+	+	+

Note: May not have addressed all potential sources of bias; –, significant sources of bias may persist; ++, the study has been designed or conducted in such way as to minimize the risk of bias; +, not clear from the way the study is reported, or that the study.

criteria and participant recruitment procedures was frequently incomplete, limiting the ability to fully evaluate methodological rigour across studies.

4.3 | Main Findings

4.3.1 | Parenting Practices Typically Associated With ADHD

The meta-analysis revealed a small and non-significant difference between ADHD+ADs and ADHD-only groups in the use of parenting practices typically associated with ADHD ($k = 2$, $g = 0.27$, 95% CI = [−0.16, 0.70], $Z = 1.24$, $p = 0.26$). There was no significant heterogeneity in effect sizes between samples ($Q_{(1)} = 0.16$, $p = 0.69$). See Figure 2.

Kepley and Ostrander (2007) found no differences between groups in conflict scores, whereas Curchack-Lichtin (2013) reported no differences in punitive parenting, inconsistent parenting and parent negative affect. Pfiffner and McBurnett (2006) also reported no significant differences in inconsistent discipline, poor monitoring/supervision and power assertion. Finally, McGee (2007) found that poor monitoring/supervision, inconsistent discipline and corporal punishment were not associated with the presence of comorbid anxiety disorders in children with ADHD. However, negative parenting significantly predicted child anxiety symptoms, accounting for up to 6% of the variance.

4.3.2 | Parenting Practices Typically Associated With Anxiety Disorders

The meta-analysis revealed a small to moderate increase in parenting practices typically associated with anxiety disorders among parents of children with ADHD+ADs compared to those of children with ADHD alone ($k = 3$, $g = 0.32$, 95% CI = [0.11, 0.53], $Z = 3.03$, $p = 0.002$). No significant heterogeneity was observed across studies ($Q_{(2)} = 0.18$, $p = 0.91$). See Figure 3.

Pfiffner and McBurnett (2006) reported that possessiveness was significantly higher among parents of children with ADHD+ADs compared to those with ADHD alone. Similarly, Kepley and Ostrander (2007) found that independence scores were significantly lower in the ADHD+ADs group. However, in this study, no significant differences were observed between groups on parental control, organisation or achievement orientation. In line with these findings, Curchack-Lichtin (2013) reported that parental control did not vary between groups.

4.3.3 | Positive Parenting Practices

Results of the meta-analysis indicated that parents of children with ADHD+ADs reported fewer positive parenting practices compared to those of children with ADHD alone ($k = 3$, $g = -0.35$, 95% CI = [−0.55, −0.14], $Z = -3.30$, $p = 0.001$). The combined effect size was small to moderate, and no significant heterogeneity was found across studies ($Q_{(2)} = 0.41$, $p = 0.81$). See Figure 4.

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Study Name	Outcome	Hedge's *g*	Std. Error	Variance	95% Confidence Interval Lower	95% Confidence Interval Upper	Z	p
	FES-Conflict	0.33	0.260	0.068	-0.18	0.84	1.253	0.210
Curchack-Lichtin (2013)	Combined	0.14	0.395	0.156	-0.65	0.91	0.353	0.724
Overall		**0.27**	**0.17**	**0.047**	**-0.16**	**0.70**	**1.241**	**0.215**

Heterogeneity: $Tau^2 = 0$; $H^2 = 1.00$; $I^2 = 0\%$

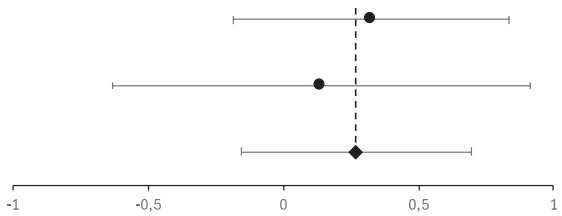


FIGURE 2 | Parenting practices typically associated with ADHD: results of the meta-analysis.

Study Name	Outcome	Hedge's *g*	Std. Error	Variance	95% Confidence Interval Lower	95% Confidence Interval Upper	Z	p
Kepley & Ostander (2007)	Combined	0.41	0.262	0.069	-0.10	0.92	1.562	0.118
Curchack-Lichtin (2013)	Combined	0.39	0.431	0.186	-0.46	1.24	0.903	0.366
	PCRQ - Possessiveness	0.30	0.120	0.014	0.06	0.53	2.465	0.014
Overall		**0.32**	**0.106**	**0.011**	**0.11**	**0.53**	**3.025**	**0.002**

Heterogeneity: $Tau^2 = 0$; $H^2 = 1.00$; $I^2 = 0\%$

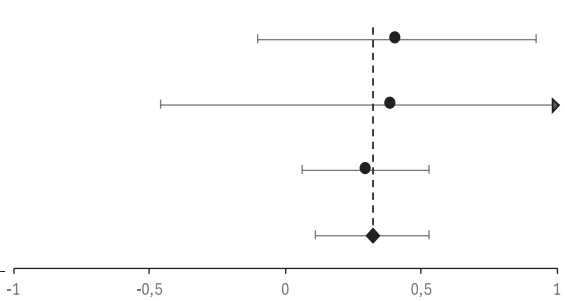


FIGURE 3 | Parenting practices typically associated with anxiety disorders: results of the meta-analysis.

Study Name	Outcome	Hedge's *g*	Std. Error	Variance	95% Confidence Interval Lower	95% Confidence Interval Upper	Z	p
	Combined	-0.30	0.262	0.068	-0.81	0.21	-1.152	0.249
Curchack-Lichtin (2013)	Combined	-0.12	0.404	0.163	-0.91	0.67	-0.297	0.767
	APQ – Positive Parenting	-0.38	0.120	0.014	-0.61	-0.14	-3.140	0.002
Overall		**-0.35**	**0.105**	**0.011**	**-0.55**	**-0.14**	**-3.296**	**0.001**

Heterogeneity: $Tau^2 = 0$; $H^2 = 1.00$; $I^2 = 0\%$

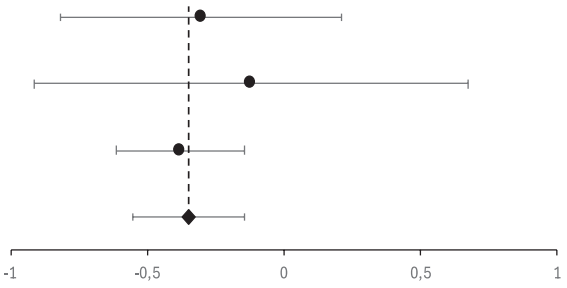


FIGURE 4 | Positive parenting: results from the meta-analysis.

According to Pfiffner and McBurnett (2006), parents of children with ADHD+ADs reported using positive parenting practices less frequently than those of children with ADHD alone. However, no significant differences were found in parental involvement or warmth scores in that study. Kepley and Ostrander (2007) observed lower active-recreational orientation scores in the ADHD+ADs group compared to the ADHD-only group, but no significant differences in communication/involvement. Curchack-Lichtin (2013) did not report any between-group differences in positive parenting practices. Finally, McGee (2007) found that positive parenting practices were not significantly associated with the presence of ADHD+ADs.

5 | Discussion

This systematic review synthesizes evidence on parenting practices in families of school-age children with ADHD+ADs compared to those with either disorder alone. Although no significant differences were found in ADHD-related parenting practices, small to moderate differences were observed in anxiety-related parenting practices and in the use of positive parenting between the ADHD+AD group and the ADHD-only group.

These findings suggest that differences in how parents interact with children with ADHD+ADs may be reflected less in

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ADHD-related practices and more in reduced positive interactions combined with increased control or overprotection. This observation is consistent with previous research highlighting the role of these anxiety-related parenting practices in the development and maintenance of anxiety disorders in children (Drake and Ginsburg 2012; Laurin et al. 2015; McLeod et al. 2007; Murray et al. 2009; Negreiros and Miller 2014; van der Bruggen et al. 2008; Vasey et al. 2014). Such practices notably encourage children to avoid confronting their fears and, consequently, contribute to the maintenance or exacerbation of anxiety (Affrunti and Ginsburg 2012; van der Bruggen et al. 2008; Van der Sluis et al. 2015). Although the effect size of anxiety-related practices was small to moderate, it is likely to be clinically meaningful, considering that the comparison group also faced the parenting challenges associated with ADHD.

The small to moderate reduction in positive parenting practices among parents of children with ADHD+ADs compared to those of children with ADHD alone is also consistent with previous research (Yap and Jorm 2015). This reduction may reflect the increased burden experienced by these families, which, in turn, can disrupt family functioning (Bowen et al. 2008; Mikami et al. 2011; Shroff and Sanchez-Lacay 2018; Souza et al. 2005; Tannock et al. 1995). Moreover, lower levels of positive parenting practices in children with ADHD at ages 3 and 5 have been prospectively associated with a higher likelihood of an anxiety disorder diagnosis at age 8 (Curchack-Lichtin 2013). Taken together, these findings suggest that low levels of positive parenting practices may constitute a risk marker for anxiety disorders in children already vulnerable due to ADHD (Gair et al. 2021; Shroff and Sanchez-Lacay 2018). This hypothesis is further supported by evidence demonstrating the protective effect of positive parenting on the development of internalizing and externalizing symptoms in children (Clayborne et al. 2021; Haack et al. 2016; McRae et al. 2020).

Overall, this systematic review and meta-analysis identified a parenting profile marked by increased use of anxiety-related parenting practices and reduced use of positive parenting practices in families of children with ADHD+ADs, compared to families of children with ADHD only. This study also highlights the lack of research on this topic, limiting conclusions regarding the specific parenting practices associated with ADHD+ADs, and how they differ from those observed in families of children with either condition alone. In fact, no study with an anxiety disorder comparison group were identified in this systematic review. An additional priority for future research is to identify parental characteristics associated with these practices. Such data are essential to advance our understanding of parenting in the context of ADHD+ADs and to guide the development of targeted psychosocial interventions for this highly vulnerable population. In the meantime, the current findings combined with previous studies (van Steensel et al. 2024) suggest that clinicians should consider systematically assessing and addressing these parenting practices as part of a comprehensive care approach for children with ADHD+ADs.

6 | Limitations of the Study

Several limitations should be considered when interpreting the findings of this study. First, the search strategy was limited to publications in French or English, which may have excluded relevant studies published in other languages. Second, the small number of eligible studies prevented the assessment of publication bias and limited the examination of how potential sources of bias may have influenced the results. Third, reporting of participant selection and recruitment procedures was frequently incomplete, and the sample were predominantly composed of Caucasian boys from the United States, thereby limiting the generalizability of the findings. Moreover, parenting practices were assessed primarily by mothers, limiting conclusions regarding fathers' parental practices. Finally, one study reported only statistically significant results, raising concerns about the possibility of an overestimation of group differences in anxiety-related and positive parenting practices.

7 | Conclusion

This systematic review and meta-analysis suggest a parenting profile characterized by increased use of anxiety-related parenting practices and reduced use of positive parenting practices in families of children with ADHD+ADs. This review also underscores the limited research on this topic and emphasizes the need to clarify the specific nature and implications of parenting practices in families of children with ADHD+ADs. Advancing knowledge in this area is essential to inform clinical decision-making and improve the efficacy of psychosocial interventions for children with ADHD+ADs.

Author Contributions

Isabelle Denis: conceptualisation, methodology, investigation, formal analysis, supervision, funding acquisition, writing – original draft. **Laurie Côté:** conceptualisation, methodology, investigation, formal analysis, writing – original draft. **Stéphane Turcotte:** methodology, formal analysis, writing – review and editing. **Guillaume Foldes-Busque:** methodology, formal analysis, writing – review and editing.

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Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

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Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analysed in this study.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:**Search strategies.

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